

Kubernetes Namespace Stuck Terminating Runbook

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Category	kubernetes	Severity	P3 - Medium
Domain	Container Orchestration	Est. Duration	15-25 minutes
Tags	kubernetes, namespace, terminating, finalizers, stuck, cleanup		

Overview

A namespace stuck in Terminating state means Kubernetes has started deleting it but cannot complete because resources within it still have finalizers that are not being processed. The most common cause is a CRD (Custom Resource Definition) whose controller is no longer running.

Prerequisites

- kubectl with cluster-admin permissions
- jq installed for JSON manipulation
- Understanding of which CRDs/operators are installed in the cluster

Response Steps

Step 1: Confirm Namespace is Stuck

Verify the namespace is in Terminating state and check how long it has been stuck.

```
kubectl get namespaces | grep Terminating
kubectl describe namespace
```

Step 2: List Resources Still in Namespace

Find all remaining resources in the namespace that are blocking deletion.

```
kubectl api-resources --verbs=list --namespaced -o name | xargs -n1 kubectl get --show-kind --ignore-not-found -n 2>/dev/null
kubectl get all -n
```

Step 3: Check Namespace Finalizers

Inspect the namespace's finalizer list. The finalizers prevent deletion from completing.

```
kubectl get namespace -o json | jq '.spec.finalizers'
kubectl get namespace -o jsonpath='{.spec.finalizers}'
```

Step 4: Remove Resource Finalizers

For each stuck resource, remove its finalizers so Kubernetes can delete it.

```
kubectl patch -n -p '{"metadata":{"finalizers":[]}}' --type=merge
kubectl get all -n
```

NOTE: Only remove finalizers if the owning controller/operator is confirmed gone.

Step 5: Remove Namespace Finalizer via API Proxy

If the namespace itself has a kubernetes finalizer blocking deletion, patch it out via the API proxy.

```
kubectl proxy &
```

```
curl -H 'Content-Type: application/json' -X PUT --data-binary @-
http://127.0.0.1:8001/api/v1/namespaces//finalize <"}, "spec": {"finalizers": []}} EOF
```

NOTE: This is the canonical method to force-delete a stuck namespace.

Step 6: Verify Namespace is Deleted

Confirm the namespace no longer appears in the cluster.

```
kubectl get namespaces
kubectl get namespaces | grep || echo 'Namespace successfully deleted'
```

NOTE: SUCCESS: Namespace does not appear in the list.

Rollback Steps

Rollback Step 1: Recreate Namespace if Deleted by Mistake

If the namespace was deleted accidentally, recreate it and restore workloads.

```
kubectl create namespace
kubectl apply -f .yaml
```

Rollback Step 2: Reinstall CRD Controller

If finalizers were removed because the controller was missing, reinstall the operator.

```
helm install -n
kubectl get pods -n
```

Step Dependencies

Step	Depends On	Can Parallelize With
Step 1: Confirm Stuck	—	—
Step 2: List Resources	Step 1	—
Step 3: Check Finalizers	Step 1	Step 2
Step 4: Remove Resource Finalizers	Step 2, Step 3	—
Step 5: Remove Namespace Finalizer	Step 4	—
Step 6: Verify Deleted	Step 5	—

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